

Project Manager Maneuver Ammunition Systems
Product Lead Special Ammunition and Weapon Systems

GENERAL SPECIFICATION FOR
MORTAR AMMUNITION
10 February 2026

1. SCOPE

1.1 SCOPE: This specification describes the requirements and verification procedures for Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 General. In addition to documents listed in this Section, there are documents specified in Section 5 of this specification. All documents listed in this specification should use the latest revision available at the time of award.

2.2 Government documents. Not Applicable

2.3 Non-Government publications.

- a) United Nations Recommendations on the Transport of Dangerous Goods Model Regulations, 24th Edition, 2025.
- b) International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air, 2025-2026 Edition.
- c) International Maritime Organization (IMO) International Maritime Dangerous Goods Code (IMDG) 2024 Edition, Amendment 42-24.

2.4 Acquisition requirements. Acquisition documentation (base award and/or Request for Proposal (RFP)) must specify the following:

- a. Title and date of this specification
- b. Requirements for certificates of conformance for each lot or shipment of product.
- c. Requirements for age of ammunition.
- d. Requirements and provisions for submission of data as required.
- e. Requirements for acceptance criteria if different than those stated in specification.
- f. Requirements for reduction or elimination of verification procedures.
- g. Requirements and provisions for contractor and Government verification.
- h. Requirements and provisions for packaging of ammunition.
- i. Requirements and provisions for transportation of ammunition.

2.5 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Requirements specified in the contract or delivery order precede the text of this document. Nothing in this document or the contract, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Technical requirements. Mortar ammunition shall meet the requirements and verifications of this specification for the ammunition listed in ANNEX A. Any ammunition that is categorized as mortar ammunition, but is not specifically identified in ANNEX A, must still adhere to the testing parameters of this specification. All mortar ammunition shall be accompanied with technical data (see APPENDIX) and meet or exceed the requirements outlined in this specification.

3.1.1 Compliance to technical requirements and storage history records. A Technical Data Package (TDP) as well as lot history folders shall be provided to ensure ammunition is in compliance with this specification. Additionally, documented storage history of the ammunition shall be provided.

3.2 Conformance Testing and Inspection. A random sample of the ammunition/items from each lot presented shall be selected by a U.S. Government (USG) Representative and subjected to conformance testing and inspection in accordance with TABLE I (to include retest rounds, if required).

3.2.1 Stock Ammunition. All stock mortar ammunition shall be less than 5 years old and meet the verification procedures in TABLE I of this specification. Any lot that has not been adequately tested to comply with all requirements of Table I shall be subjected to conformance testing and inspection in accordance with this specification.

3.3 Serviceability. Mortar ammunition shall be serviceable/issuable, in good condition and without visible signs of degradation. All parts and assemblies shall be fabricated and finished in a thorough manner and shall be free of burrs, chips, dirt, grease, rust, corrosion, and other foreign material that adversely affect fit or function. Coatings and protective finishes shall be uniform and shall provide complete coverage of designated surfaces. The painting and marking shall be restored by the assembling facility on any components that have been marred or obliterated.

3.3.1 Identification of defects. Visual defects shall be identified within the TDP and include a defined method for acceptance/rejection of mortar ammunition. Visual inspections listed in ANNEX C shall be the minimum requirement.

3.4 Weapon interface and interoperability. The mortar ammunition shall conform to weapon interface as identified in ANNEX A or as otherwise specified in contract or delivery order.

3.5 Operating requirements. The mortar ammunition shall provide their functional, operational, and performance capabilities when fired from the weapon systems listed in ANNEX A regardless of weather related conditions (example: rain, snow, mud, wind, fog). If testing occurs in inclement weather, the Contractor shall implement all necessary safety precautions.

3.5.1 Operating temperatures. The mortar ammunition shall demonstrate safety and reliability throughout exposure to ambient and extreme temperatures. Mortar ammunition shall reliably perform within a temperature range of -30°C to +50°C.

3.5.2 Muzzle velocity. The average velocity obtained during LAT at ambient temperature testing shall meet the specified characteristics listed in ANNEX B. Each lot shall meet the standard deviation requirements. The value of the deviation if not provided in Request for Proposal (RFP) will be determined during TDP development.

3.5.3 Chamber pressure. The mortar ammunition shall meet the maximum individual chamber pressure characteristics at ambient specified in ANNEX B. It shall also meet the average pressure at ambient and maximum individual pressure requirements at hot specified in the TDP.

3.5.4 Maximum effective range. The maximum corrected range of the mortar ammunition shall meet or exceed at ambient temperature the specified requirement listed in ANNEX B.

3.5.5 Dispersion. The mortar ammunition shall demonstrate dispersion at ambient temperature as specified by the manufacturer in the TDP.

3.5.6 Function and casualty. The mortar ammunition shall function without casualty or cause damage to the weapon and shall exhibit a ballistic performance free of misfires, duds, premature bursts, short rounds, and metal parts separation. The ammunition shall cause no hazard or unsafe conditions to the user during storage, transportation or use.

3.5.7 Fragmentation. The HE mortar ammunition shall provide the lethal fragmentations at the desired radius for each item as identified in ANNEX B.

3.5.8 Illumination. The Illumination mortar ammunition shall provide the candlepower illumination intensity as identified in ANNEX B and parachute suspended burn time as specified by the manufacturer in the TDP.

3.5.9 Smoke. The Smoke mortar ammunition shall produce smoke upon impact with the target as specified by the manufacturer in the TDP.

3.5.10 Airburst. The Airburst mortar ammunition shall provide the lethal radius and burst height as specified by the manufacturer in the TDP.

3.6 Fuze identification and description. Within the TDP, the fuze shall be identified with a drawing/figure of the internal fuze design with components categorized to include the type and weight of energetic material(s) and a functional description of the fuze that includes, at a minimum, a detailed description of the fuze arming sequence, operating temperatures, in-process tests (example: jolt/jumble, sensitivity, safe arming/non –arming etc.), and no arm and all arm distances. Test records shall be available to the USG for verification. Fuze shall be set to safest setting, if applicable.

3.7 Environmental requirements.

3.7.1 Sequential rough handling. Cartridges shall be capable of withstanding the rigors of the sequential rough handling and transportation throughout extreme temperature ranges and meet all performance and safety requirements.

3.7.2 Shelf life. The mortar ammunition shall meet all requirements of this specification within its operating temperature range at a minimum of ten (10) years of storage. All cartridges are required to reliably be stored within a temperature range of -30°C to +50°C at a minimum of ten (10) years.

4. VERIFICATION

4.1 Technical verification. The cartridges shall be subjected to verification of all requirements cited in Section 3 in this specification, in accordance with TABLE I. Noncompliance to any requirements shall be cause to withhold acceptance of the lot or batch in which the noncompliance was found.

TABLE I. Requirement/Verification Cross-Reference Matrix

Section 3 Requirements	Requirement	Section 4 Verification Procedures	Sample Size
3.1	Technical requirements	4.1	-
3.3	Serviceability	4.3	Note 1
3.3.1	Identification of defects	4.3	ANNEX C
3.4	Weapon interface and interoperability	4.4	ANNEX A
3.5.1	Operating temperatures	4.5.1	ANNEX A
3.5.2	Muzzle velocity	4.5.2	ANNEX B
3.5.3	Chamber Pressure	4.5.3	ANNEX B
3.5.4	Maximum effective range	4.5.4	ANNEX B
3.5.5	Dispersion	4.5.5	ANNEX B
3.5.6	Function and casualty	4.5.6	ANNEX A
3.5.7	Fragmentation	4.5.7	ANNEX A
3.5.8	Illumination	4.5.7	ANNEX A
3.5.9	Smoke	4.5.7	ANNEX A
3.5.10	Airburst	4.5.7	ANNEX A
5.1	Packaging	-	ANNEX C
5.3	Packing	-	ANNEX C
5.5	Marking	-	ANNEX C
Note 1: To be performed using a defined sampling procedure for inspection determined by the mortar cartridge lot sizes.			

4.1.1 Lot formation. The mortar ammunition shall be assembled into identifiable lots, sublots, or batches, or in such other manner contingent on approval by USG. Each lot or batch shall, as far as

practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time. Lot quantity shall be determined by the total quantity for shipment and include the additional rounds from that same lot to be consumed during lot acceptance testing along with contingency rounds for potential re-testing. The lots or batches shall be identified by the contractor and shall be kept intact in adequate and suitable storage space. All cartridges in a given lot must be manufactured prior to lot acceptance testing.

4.1.1.1. Lot Size. All lot sizes are to remain consistent throughout an awarded delivery order. Once temperature testing is conducted, all skip lot sizes shall be within 20% of that lot's size. If a lot size larger than 20% is proposed, temperature testing has to be conducted on that lot. If a lot fails for any reason, the next two lots presented for additional testing have to be within 20% of the failed lot size.

4.1.1.2 Component Lots. If multiple lots of a component are present, a minimum of 10 rounds shall be fired for each lot.

4.1.2 Retest criteria. A ballistic retest is authorized upon initial failure of mortar ammunition to meet test requirements as long as the failure is not considered critical. Retest procedures must be clearly defined in the TDP and results must be combined with data from the first test unless a failure analysis confirms that the failure was caused by something other than the ammunition (example: weapon failure or faulty test set up).

4.2 Verification of compliance. Evidence shall include, but is not limited to, the following;

- a. Identification of technical requirements to which the mortar ammunition is/was produced.
- b. Manufacturer, date of manufacture and original acceptance documentation.
- c. In-Process acceptance reports, including type, lot or batch identification, quantity and method of acceptance (example: sample size, verification method, acceptance criteria, results).
- d. Surveillance reports, including lot or batch identification, quantity and method of surveillance (example: sample size, verification method, criteria for action to be taken on lot or batch, results).
- e. Storage history, including duration and storage condition (example: controlled, uncontrolled).

4.3 Serviceability and identification of defects. A random sample shall be selected from each lot of mortar ammunition for inspection. The ammunition shall be visually inspected in accordance with ANNEX C of this specification. Reinspection of a lot that fails for non-ballistic reasons is authorized as long as the criteria for reinspection is clearly outlined in the TDP per ANNEX C.

4.4 Weapon interface and interoperability. A random sample shall be selected from each lot of mortar ammunition to be delivered and inserted into a weapon chamber or chamber gage that conforms to the identified weapon system to check for profile and alignment. Inability to interface properly with weapon system shall be considered a failure to show compliance and constitute a single reliability failure.

4.5 Operating verification. The same cartridge of mortar ammunition can be used for multiple tests provided the test equipment is capable of capturing all necessary test data. For example, Muzzle Velocity, Maximum Effective Range and Function & Causality testing can be done at the same time using the same mortar cartridge. Called warmers and spotters are allowed and encouraged. "Called" means that before the mortar cartridge is loaded into the weapon, the USG representative must be informed that the next mortar cartridge to be fired is a warmer or spotter. Should a mortar cartridge not be "Called" it shall be counted toward whichever test is being conducted, potentially resulting in a LAT failure. Critical safety failures apply to all cartridges fired during LAT, regardless of which test is being performed. The following safety failures shall result in the immediate rejection of the lot; there shall be no retest:

- a. Detonation, Deflagration, or function within the weapon (all cartridge types).
- b. Detonation of a High Explosive (HE) or High-Explosive Anti-Tank (HEAT) cartridge before impact with the ground or target.
- c. Any early functions of an Illumination cartridge within 2 seconds of launch.
- d. Any early function of an Airburst cartridge at a range of 80% or less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
- e. An early function of a Practice (Live) cartridge before impact with the ground or target which exposes the operator to smoke, fragmentation or fire.
- f. An early function of a Smoke, Airburst, or Illumination cartridge which exposes the operator to smoke, fragmentation or fire.
- g. Metal parts separation.
- h. Short rounds: a cartridge impacts the ground at a range of 80% or less of the range average or its respective temperature/charge/group; excluding the cartridge(s) in question.
- i. A cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools).

4.5.1 Operating temperatures. Mortar ammunition shall operate reliably and safely when conditioned and fired throughout a reasonable extreme operating temperature range. Mortar ammunition shall consistently perform within a temperature range of -30°C to +50°C, ensuring reliable function under the specified environmental conditions. Temperature testing may be conducted on a skip-lot basis (see APPENDIX) during LAT. A single safety failure will result in lot rejection and there shall be no retest. Functional testing parameters must be defined in the technical data and meet manufacturer requirements. The entire mortar cartridge, including all components necessary to fire and function (example: fuze, warhead, propellant, etc.) shall be temperature conditioned based on the maximum mortar cartridge body diameter. Additionally, the mortar cartridges shall be fired within the following times after being removed from the conditioning chamber or thermally insulated bag.

- a. Ammunition less than 85mm in diameter shall be conditioned for not less than six (6) hours and fired within five (5) minutes of removal from conditioning box or thermally insulated bag.
- b. Ammunition between 85mm and 120mm in diameter shall be conditioned for not less than nine (9) hours and fired within six (6) minutes of removal from conditioning box or thermally insulated bag.
- c. Ammunition greater than 120mm in diameter shall be conditioned for not less than twelve (12) hours and fired within eight (8) minutes of removal from conditioning box or thermally insulated bag.

4.5.1.1 Propellant charge operational test criteria: The minimum propellant charge zone shall be used at the lowest operating temperature and the maximum propellant charge zone shall be used at the highest operating temperature.

4.5.2 Muzzle Velocity. Velocity tests shall be conducted in accordance with the manufacturer's test procedures to verify that a sample representative of the lot meets the minimum velocity requirements specified in ANNEX B. If the lot fails to meet the Velocity requirements a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all mortar cartridges tested (20 cartridges minimum) shall be used to determine if the lot passes. Each lot shall meet the standard deviation requirements specified in the RFP or TDP.

4.5.3 Chamber Pressure. Pressure tests shall be conducted in accordance with the manufacturer's test procedures to verify samples from the lot do not exceed maximum individual peak pressure requirements listed in ANNEX B (10 cartridges minimum), the average pressure at ambient listed in the TDP (10 cartridges minimum), and the maximum individual pressures at hot listed in the TDP (10 cartridges minimum). Each lot of propellant charges shall meet the individual maximum and average pressure requirements. Pressure testing shall be conducted during the LAT, regardless of in-process testing. Any single round that exceeds the maximum individual peak pressure requirement will result in lot rejection, there shall be no retest.

4.5.4 Maximum Effective Range. Maximum effective range shall be verified during lot acceptance testing to ensure compliance with ANNEX B.

4.5.5 Dispersion. Dispersion tests shall be conducted to verify that each lot of ammunition meets the specified requirements in the TDP (10 cartridges minimum). If the lot fails to meet the Dispersion requirements a retest of the same sample size (10 cartridges minimum) is permitted. The cumulative average of all cartridges tested (20 cartridges minimum or 19 cartridges if one misses the target), shall be used to determine if the lot passes. If two (2) or more cartridges miss the target (cumulative) the lot fails.

4.5.6 Function and casualty. A random sample shall be selected from the ammunition lot in accordance with ANNEX A to be delivered and functioned from the identified weapon system to check for function, casualty and metal parts security. The ammunition shall function across the entire operating temperature range (extreme hot, cold, ambient) without casualty or damage to the weapon, and shall exhibit a ballistic performance free of misfires, duds, premature bursts, short rounds, and metal parts separation. Critical Safety failures are listed in Section 4.5 of this specification. The ammunition shall cause no hazard or unsafe condition to any personnel during storage, transportation, or use. Any rounds which detonates prior to impact of intended target, or which detonates in the weapon is considered a safety critical failure and will result in immediate lot rejection. If a cartridge has a multi-option fuze, all options shall be tested during Lot Acceptance Testing to properly demonstrate the fullest extent of the munitions capability unless specified otherwise in Annex A or by the USG during solicitation. The different fuze options test quantity will be distributed amongst the total Function and Casualty requirement. The ammunition shall cause no hazard or unsafe conditions to the user during storage, transportation or use. Critical safety failures are listed in Section 4.5 of this specification.

4.5.7 Performance. Proper functioning for each mortar ammunition type is defined below. Any round which does not function properly according to the requirements of the manufacturer's test procedures and the definitions below are considered a function failure.

- a. Fragmentation: The ammunition, with a point detonate fuze, shall produce a high-order detonation of the warhead upon initial impact with the target (terrain or man-made structure) as specified in the TDP. If fuze contains a delay the ammunition shall detonate after impact with target at a specific time or distance as specified in the TDP.
- b. Smoke: The ammunition shall produce a volume of smoke upon impact with the target as specified by the manufacturer in the TDP.
- c. Illumination: The ammunition shall produce parachute suspended light at specific wavelength and height as specified by the manufacturer in the TDP. Intensity and duration of the parachute suspended light shall be as per ANNEX B of this specification.
- d. Airburst: The ammunition shall produce a high-order detonation of the warhead above target between a min/max distance specified by the manufacturer in the TDP.
- e. Practice (Inert): The ammunition shall not contain any explosives or energetics other than propellant.
- f. Practice (Live): The ammunition may contain energetics, other than propellant, to signal location of impact.

5. PACKAGING

5.1 Preservation, packaging, packing, unitization, and marking shall provide protection for multiple handling, redistribution, and shipment by any transportation mode and meet or exceed the following requirements.

5.1.1 Packaging containers for hazardous materials, ammunition and explosives shall meet or exceed the requirements found in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.1.2 Unit Package – The unit package shall be designed and constructed so that it will contain the contents with no damage to the item(s), with minimal damage to the unit pack during shipment and storage in the shipping container, and will allow subsequent handling. The outermost component of the unit package shall be a container such as a sealed tube, carton or box.

5.1.3 Cleanliness – Items and packaging shall be free of dirt and other contaminants which would contribute to the deterioration of the item or which would require cleaning by the customer prior to use. Coatings and preservatives applied to the item for protection are not considered contaminants.

5.1.4 Preservation – Items susceptible to corrosion or deterioration shall be provided protection against external environmental effects.

5.1.5 Cushioning – Items requiring protection from physical and mechanical damage (e.g. fragile, sensitive, critical material) or which could cause physical damage to other items, shall be protected by wrapping, cushioning, pack compartmentalization, or other means to mitigate shock and vibration and prevent damage during handling and shipment.

5.1.6 Wood Packing Material (WPM) – All-natural wood packing material shall be constructed from heat treated lumber and certified by an accredited recognized agency in accordance with International Standards for Phytosanitary Measures (ISPM-15) regardless of origin or destination. All natural wood packing materials must include certification markings that are readily visible to inspectors and on at least two sides of every container when palletized.

5.1.7 Firing Tables – A copy of the manufacturer's firing tables for ammunition shall be included with the TDP.

5.2. Packing

5.2.1 All shipping containers shall be the most cost effective and shall be of the minimum cube to contain and protect the items.

5.2.2 Shipping Containers – the shipping container (including any necessary blocking, bracing, cushioning, or waterproofing) shall comply with the regulations of the carrier used and shall provide safe delivery to the destination at the lowest tariff cost. The shipping container shall be capable of multiple handling, stacking at least ten feet high, and storage under favorable conditions and meet the requirements of the "United Nations Recommendations on the Transport of Dangerous Goods".

5.3. Unitization

5.3.1 Shipments of identical items going to the same destination shall be palletized if they have a total cubic displacement of 20 cubic feet or more unless skids or other forklift handling features are included on the containers. Pallet loads must be stable, and to the greatest extent possible, provide a level top for ease of stacking. The weight capacity of the pallet must be adequate for the load. A pallet load shall not exceed 4,000 pounds and should not exceed 52 inches in length or width, or 54 inches in height. The load shall be contained in a manner that will permit safe handling during shipment and storage.

5.3.2 Banding – Metal banding shall be used to secure the load. Straps shall be applied to each column or layer of boxes. Tie down straps shall be applied to each column of boxes at 90 degrees to the load straps. Edge protectors shall be used when securing fiberboard boxes.

5.4. Marking

5.4.1 Packaging marking shall be visible, clear, and remain legible during normal life cycle handling.

5.4.2 All unit packages, intermediate packs, exterior shipping containers, and, as applicable, unitized loads shall be marked with item description, quantity, lot number, or serial number. The outer shipping container and unitized load shall indicate load weight, UN dangerous goods proper shipping name, and location of UN number and UN POP Marking.

5.4.3 Each package shipping container shall show the United Nations packaging symbol and applicable codes in accordance with the construction requirements and testing of packaging as expressed in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations".

5.5. Additional Requirements for Hazardous Materials

5.5.1 The shipment shall be prepared in accordance with the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and other applicable regulations effective at the time of shipment in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.5.2 Packaging and marking for hazardous material shall comply with the requirements for the mode of transport and the applicable performance packaging contained in the following documents:

- a. International air transport: International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air
- b. International vessel transport: International Maritime Organization (IMO) International Maritime Dangerous Goods Code (IMDG)

6. REPORTING

6.1 Lot Acceptance Test Reports (LATR), at a minimum, must include a summary of findings for all visual and ballistic testing, lot history folders (in-process testing per TDP requirements, ammo data card, and all certifications or certificates), all measured data, pressure (at ambient, hot) and velocity for each individual shot, range for all test shots, deflection for all test shots scored for dispersion, identify conditioning temperature used, calculated average pressure and velocity, spread for minimum and maximum velocity, and calculated dispersion. The LATR must also include the ammo data card for all lots tested.

6.1.1 Ammo Data Cards are required to be provided as part of the LATR and must include lot information for all energetics and major components used in the production of each lot.

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Annex A: Weapon Interface
Annex B: Operating Requirements
Annex C: Visual Inspection
APPENDIX - Definitions

ANNEX A – Weapon Interface

Function & Casualty for order quantities of 1000+ mortar cartridges ⁴				
Canon Compatibility ^{1,2}	Item ^{8,9}	Testing		
		Sample Size-Accept-Reject & Retest (cumulative) ³		
60mm Commando 60mm: M57, M6-211LR 82mm: 2B14, M69A 120mm: 2B11, 2S12, M75	High Explosive ^{5,6} Practice ⁵ Smoke ⁵ Airburst ⁷	Ambient	Hot	Cold
		15-0-1 Retest 29-1-2 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)	15-1-2 Retest 29-2-3 (Cumulative)
	Illumination	15-2-3 Retest 29-4-5 (Cumulative)	15-2-3 Retest 29-4-5 (Cumulative)	15-2-3 Retest 29-4-5 (Cumulative)
Notes on next page				

Function & Casualty for order quantities of 100 – 999 mortar cartridges ⁴		
Canon Compatibility ^{1,2}	Item ^{8,9}	Testing
		Sample Size-Accept-Reject & Retest (cumulative) ³
60mm Commando 60mm: M57, M6-211LR 82mm: 2B14, M69A 120mm: 2B11, 2S12, M75	High Explosive ^{5,6} Practice ⁵ Smoke ⁵ Airburst ⁷	Ambient only
		15-0-1 Retest 29-1-2 (Cumulative)
	Illumination	15-2-3 Retest 29-4-5 (Cumulative)
Notes on next page		

ANNEX A – Weapon Interface (continued)

Note 1: Any mortar ammunition not listed in this table is still required to meet sample size-accept-reject criteria listed in this table.

Note 2: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved by the USG.

Note 3: These samples sizes shall be used to show performance and function and casualty at ambient temperature. 15 rounds shall be tested for function and safety at the operating temperature extremes. A minimum of 10 data points is required to evaluate all other test parameters.

Note 4: For orders less than 100 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Note 5: If the fuze has an adjustable time delay setting it must be set as per approved TDP during Function & Casualty testing.

Note 6: For these cartridge types only, if they contain a multi-option fuze where the only options are Point Detonate and Delay, they shall be set to Point Detonate during LAT.

Note 7: For Airburst cartridges, the range, lethal radius as well as height of burst shall be provided within the TDP.

Note 8: A function failure shall be defined as:

- a) A cartridge that will not properly load into the weapon.
- b) A misfire - failure of a projectile to discharge as intended.
- c) HE cartridges only: A cartridge that does not completely detonate upon initial impact with their target when equipped with a point detonate fuze. Self-destruct detonation or fuze function only shall be scored as a function failure, if shot as part of a function test. If fuze is set to delay the ammunition shall detonate after impact with target at a specific time or distance as specified in the TDP.
- d) Smoke cartridges only: A cartridge that does not produce specified volume of smoke or has early function that does not expose user to smoke, blast, fire, or fragmentation.
- e) Illumination cartridges only: A cartridge that does not produce light for specified intensity, wavelength, duration and height or has early function that does not expose user to smoke, blast, fire, or fragmentation.
- f) Practice (Live) cartridges only: A cartridge that has an early function, or a failure to function before impact with the ground or target that does not expose user to smoke, blast, fire, or fragmentation.
- g) Airburst cartridges only: A cartridge that does not detonate at the designated height above target or has early function that does not expose user to smoke, blast, fire, or fragmentation.

Note 9: Safety failures are listed in Section 4.5 of this specification.

ANNEX B – Operating Requirements

Mortar Ammunition						
Item ^{1,3,5}	Average Velocity (m/s)	Maximum Individual Chamber Pressure (kgf/cm ²) at Ambient	Maximum Range (m)	Lethal Fragmentation Radius ² (m)	Illumination ⁴	
					Candlepower (Cd)	Duration (sec)
60mm Commando	≥ 130	≤ 442	≥ 1000	≥ 10	≥ 150,000	≥ 20
60mm	≥ 190	≤ 505	≥ 1500	≥ 10	≥ 150,000	≥ 20
82mm	≥ 205	≤ 660	≥ 3000	≥ 15	≥ 500,000	≥ 30
120mm	≥ 270	≤ 987	≥ 5500	≥ 18	≥ 1,000,000	≥ 40
Note 1: A minimum of 10 data points shall be required for velocity, pressure, and dispersion. Pressure values are based on copper crusher method. Multiple tests can be combined to reduce number of total cartridges fired. Example: the same cartridge may be used to measure Pressure, Velocity, and dispersion provided the equipment is capable of taking all measurements at the same time. Note 2: Lethal fragmentation radius applies to all fragmentation cartridges such as HE and Airburst. Note 3: For Practice cartridges: Velocity and Effective Range shall be per the manufacturer's requirements unless specified by the USG in the proposals. Note 4: Candlepower applies to all Illumination cartridges. Note 5: Safety failures are listed in Section 4.5 of this specification.						

ANNEX C – Visual Inspection Requirements

Critical: 24-0-1	Major: 24-1-2 Re-inspection: 72-3-4 (cumulative)	Minor: 24-3-4 Re-inspection: 72-9-10 (cumulative)
Missing, cracked, loose or bent parts	Cases with moving cartridges; loose pack	Discolored, oily, dirty, or smeared
Gross Corrosion / Pitting	Incorrect quantity of cartridges in packaging (light boxes shall have their corners painted) each cartridge missing shall constitute one failure	Minor surface corrosion
Spillage/leakage of energetic materials	Wire Security Seal missing	Dents, chips or burrs
Incorrect Marking of cartridge type: HE or Airburst cartridge marked as Practice, Smoke or Illumination	Mixed ammunition types: Training cartridge mixed with Combat cartridge lot	Scratches on cartridge
Gross deformation	Missing, illegible or incorrect marking	Excessive grease on cartridge
Mixed ammunition types: Combat cartridge mixed with training cartridge lot	Incorrect Marking of cartridge type: Practice, Smoke or Illumination cartridges marked incorrectly	Irregular disposition of cartridges in case
Bourrelet diameter too large	Primer missing, inverted, double, cocked, or visually above flush	Paint damaged, wet or incomplete
Note 1: If the number of allowable defects is exceeded a double sample quantity shall be reinspected. The accept/reject criteria is based on the cumulative number of defects for all three samples. Note 2: If a critical is found or the number of allowable majors or minors, after reinspection, is exceeded the lot fails. Note 3: The inspection and reinspection quantity listed is selected during LAT. DCMA will utilize their Zero-Based Acceptance Sample Plans shown in Appendix during GSI.		

APPENDIX

1. Definitions.

1.1 PdL SAWS Mortar Ammunition. PdL SAWS mortar ammunition are munitions that have not been safety tested and type classified for Army use, and cannot be procured through the US Army supply. Munitions and explosives that are not managed by National Inventory Control Points, have not been safety tested nor type classified for US Army use, do not have a national stock number (NSN) and cannot be procured or requisitioned through the Army or other Department of Defense supply system.

1.2 Technical data. Technical data, otherwise known as the Technical Data Package (TDP) is the product specific technical drawing and Quality Assurance (QA) requirements to which ammunition and associated packaging is produced and accepted for each applicable Contract Line Item Number (CLIN). For the purpose of this specification, the TDP, as a minimum, must contain the following: The Government Contract Number and Delivery Order; top assembly, fuze and key component drawings with revision number, revision date, document control signature, key interface dimensions, and a list of component assemblies and drawings and revision dates; packaging and marking drawings with revision number, revision date, and markings; and product specification with visual, dimensional and ballistic acceptance test methods, including in-process testing, with sample size, accept/reject criteria, and retest criteria. Within the TDP there shall also be a clear cut-away drawing of the fuze to show all the inner parts and workings along with a paragraph to describe the fuze arming sequence and events leading to function. A copy of the manufacturer's firing tables shall be included with the TDP.

1.3 Degradation. Ammunition with gross nonconformance to identified technical requirements, corrosion, cracks, deformation, and spillage.

1.4 Deterioration. Packaging that is ripped, broken, perforated, water damaged, or crushed.

1.5 Skip-Lot Temperature Testing. Testing at the operational temperature limits shall be conducted on the first lot of every delivery order and every 5th lot thereafter (1, 5, 10, 15...). The ammunition must pass the requirements defined in ANNEX A. If a lot is rejected for any reason, the following two lots must pass extreme temperature testing before resuming the skip lot procedures.

APPENDIX (continued)

	Acceptable Quality Level (AQL)															
LOT SIZE	0.01	0.015	0.025	0.04	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10.0
1-8	A	A	A	A	A	A	A	A	A	A	A	A	5	3	3	3
9-15	A	A	A	A	A	A	A	A	A	A	13	8	5	3	3	3
16-25	A	A	A	A	A	A	A	A	A	20	13	8	5	3	3	3
26-50	A	A	A	A	A	A	A	A	32	20	13	8	7	7	5	3
51-90	A	A	A	A	A	A	80	50	32	20	13	13	11	8	5	4
91-150	A	A	A	A	A	125	80	50	32	20	19	19	11	9	6	5
151-280	A	A	A	A	200	125	80	50	32	29	29	19	13	10	7	6
281-500	A	A	A	315	200	125	80	50	48	47	29	21	16	11	9	7
501-1200	A	800	500	315	200	125	80	75	73	47	34	27	19	15	11	8
1201-3200	1250	800	500	315	200	125	120	116	73	53	42	35	23	18	13	9
3201-10,000	1250	800	500	315	200	192	189	116	86	68	50	38	29	22	15	9
10,001-35,000	1250	800	500	315	300	294	189	135	108	77	60	46	35	29	15	9
35,001-150,000	1250	800	500	490	476	294	218	170	123	96	74	56	40	29	15	9
150,001-500,000	1250	800	750	715	476	345	270	200	156	119	90	64	40	29	15	9
500,001 & Over	1250	1200	1112	715	556	435	303	244	189	143	102	64	40	29	15	9

Note: An "A" in place of a number indicates that all units in a lot must be inspected